

JOSHUA NATHANIEL C. TATING

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EDUCATION

De La Salle University

Bachelor of Science in Computer Science, Major in Software Technology

Manila, Philippines

Aug 2022 – Dec 2026

- **Cumulative GPA:** 3.12/4.00
- **Awards:** Consistent Dean's Honors Lister
- **Organizations:** La Salle Computer Society, DLSU Viridis Arcus Esports, DLSU Green Aces MLBB Team

EXPERIENCE

Software Engineer | NestJS, Fastify, GraphQL, Next.js, Docker, Prisma, PostgreSQL

Nov 2025 – Present

NGnair Payments

Wyoming, USA

- Own **3** of the platform's core services; engineered a fault-tolerant payment orchestration service integrating **5** payment processors (Elavon, Worldpay, Deluxe) with automatic failover and activation-time routing configuration, eliminating a per-request cross-service round-trip and a class of checkout-blocking failures.
- Delivered an end-to-end integration of a new card processor solo: OAuth2 client-credentials auth with token caching, PCI-scoped hosted payment fields, merchant boarding with identity verification, full payment lifecycle (authorize, capture, refund, void), and fail-closed, idempotent webhook ingestion.
- Redesigned an **AI-assisted merchant underwriting pipeline** into a 3-stage architecture (single-call LLM risk synthesis, deterministic cost gating, code-based policy engine), cutting LLM API costs by **83%** and producing fully replayable audit records for compliance review.
- Built a public API security layer with OAuth2 client-credentials flow, scoped machine JWTs (RS256/JWKS), per-consumer API keys with constant-time validation, and fail-closed token verification enforcing per-client data boundaries.
- Migrated a centralized authentication service from Rails to NestJS, implementing JWT issuance, refresh-token rotation, MFA, SSO token exchange, and standardized RBAC scopes consumed by **9+** **microservices** through an Apollo Federation supergraph.
- Extracted checkout and merchant onboarding into independently deployable **Next.js** applications, hardening them with session-bound access control, CSP nonces, and origin-locked iframe **postMessage** embedding for secure cross-origin portal integration.

Freelance Developer | TypeScript, Express.js, Prisma, PostgreSQL, React, MongoDB, Node.js

Mar 2025 – Present

Independent Clients

Manila, Philippines

- Built a transportation booking API for a logistics company in **Express 5, TypeScript, Prisma, and PostgreSQL**, implementing a documented **25-endpoint OpenAPI** contract with **JWT** role-based auth and **Zod** validation.
- Modeled the booking lifecycle as an enforced state machine with terminal-state locking, a **24-hour** edit cutoff, and audit logging on every status change, covered by **28 Jest** integration tests.
- Engineered a full-stack e-commerce platform using the MERN stack with Material-UI, implementing JWT-based role-based access control for Admin, Staff, and Customer portals.

PROJECTS

Distributed Online Enrollment System | Go, PostgreSQL, React, TypeScript, Docker

Aug 2026

- Built a fault-isolated enrollment platform of **5 Go** services on **Docker Compose**, verifying **Ed25519 JWTs** locally so signed-in users kept working while the auth node was down.
- Prevented over-enrollment with **PostgreSQL** row-level locking, proven by integration tests racing **20 concurrent enrollments** into a **5-seat** course that admitted exactly **5**.
- Added a **PostgreSQL 17** streaming replica via **pg_basebackup** and a cursor-based notifier service that emails enrollment and grade events and resumes after restarts.

ChronoCritters | Java, Spring Boot, React, TypeScript, gRPC, GraphQL, WebSockets

Sept 2025 – Oct 2025

- Built **3 Spring Boot** microservices communicating over **gRPC**, REST, and **GraphQL**, consolidating shared models, mappers, and security config into a common library of **40** classes reused by **29** service files.
- Engineered a **STOMP-over-WebSocket** lobby service with JWT-authenticated matchmaking and per-battle topics, broadcasting turn state and turn timers to both players in real time.
- Designed a battle engine using the **Chain of Responsibility** pattern with **4** turn handlers (abilities, effects, fainting, turn transition), cutting the core battle service from **257** to **182** lines (**29%**).
- Developed a typed **React** and **TypeScript** frontend using **GraphQL Code Generator** types for **8** queries and mutations, keeping client data shapes in line with the backend schema.

Socratic AI Tutor (Thesis) | Next.js, PostgreSQL, Vercel AI SDK, LLM Tool Calling

Aug 2025 – Feb 2026

- Built an **agentic LLM tutoring system** using the Vercel AI SDK and Groq API, implementing structured tool-calling to orchestrate reasoning workflows and transition the model from open-ended generation to deterministic misconception detection.
- Designed sequential prompt engineering and evaluation pipelines with in-context reflection constraints, reducing diagnostic hallucinations (False Positive Rate) from **13.3%** to **1.3%** and achieving **98.0% precision** in detecting code misconceptions.
- Architected a **Human-in-the-Loop (HITL) LLM validation backend** using Next.js and PostgreSQL to ground responses in verified data and allow educators to govern dynamic context injection.

Multi-Core OS Process Emulator | C++20, CMake

Oct 2025 – Nov 2025

- Built a multi-core OS process emulator in **C++20** simulating up to **255 CPU cores**, running **1,000–2,000**-instruction processes under **FCFS** and **Round-Robin** scheduling with configurable time quanta.
- Parallelized each CPU tick with one **std::jthread** per busy core, synchronizing a global cycle clock with mutexes and condition variables, and interpreted **5** instruction types including nested loops.

TECHNICAL SKILLS

Languages & Databases: Java, Go, C/C++, JavaScript, TypeScript, Python, SQL, PostgreSQL, MySQL, MongoDB, Redis, HTML, CSS, Bash
Frameworks & Libraries: Spring Boot, NestJS, Next.js, React, Node.js, Express.js, Fastify, GraphQL, Apollo Federation, gRPC, Protocol Buffers, Socket.IO, Prisma, Mongoose, Zod, OpenAPI, Lombok, Zustand, Tailwind CSS, Pandas, NumPy, scikit-learn
Tools & Platforms: Docker, Docker Compose, Git, GitHub Actions, AWS S3, Linux, Nginx, CMake, Vercel, Render, Vite, Supabase, Jest, Vitest, JUnit, Playwright, Supertest, Postman, Jupyter, Neovim, VS Code, Arch Linux, Claude Code